

Math

22 QUESTIONS | 35 MINUTES

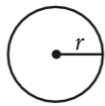
DIRECTIONS

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NOTES

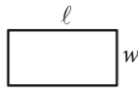
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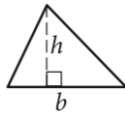


$$A = \pi r^2$$

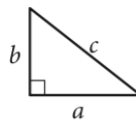
$$C = 2\pi r$$



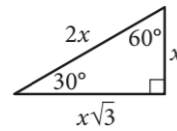
$$A = \ell w$$



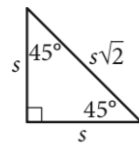
$$A = \frac{1}{2}bh$$



$$c^2 = a^2 + b^2$$



Special Right Triangles



$$V = \ell wh$$



$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}\ell wh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of the measures in degrees of the angles of a triangle is 180.

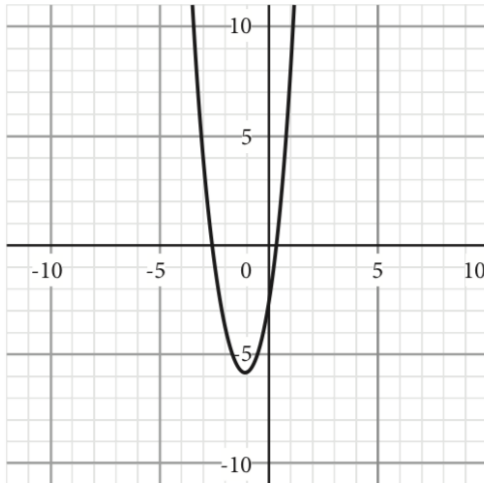
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1

Which of the following represents the equation of the graph below?



- A) $y = 3x^2 - 6x - 3$
- B) $y = -3x^2 + 6x - 3$
- C) $y = 3x^2 + 6x - 3$
- D) $y = -3x^2 - 6x - 3$

2

What is the product of the solutions in the equation below?

$$3x^2 - 12x + 15 = 0$$

- A) 3 B) 4 C) 5 D) 6

3

A standard 52-card deck consists of 13 hearts, 13 spades, 13 clubs, and 13 diamonds. If the first card picked randomly is a heart and is not put back on the deck, what is the probability of picking a second heart card from the deck?

- A) $\frac{13}{52}$
- B) $\frac{12}{52}$
- C) $\frac{12}{51}$
- D) $\frac{39}{51}$

4

Which expression is equivalent to $6xy(y + 3x) - 2x(3xy + y^2)$?

- A) $12xy^2 + 12x^2y$
- B) $4xy^2 + 12x^2y$
- C) $4xy^2 - 12x^2y$
- D) $12x^2y - 4xy^2$

5

The price of a dress, which originally cost \$200, increased by 15% on 15th May 2021. Mary bought the same dress on the 23rd of May 2021 at a 30% discount. How much did she pay for the dress?

- A) 156 B) 159 C) 161 D) 163

6

If $x^2 - y^2 = 12$ and $x - y = 2$, what is the value of $x + y$?

- A) 6
- B) 4
- C) 2
- D) -2

7

What is the value of xy for the system below?

$$2x + y = 2$$

$$3y - x = 20$$

- A) -12
- B) -2
- C) 6
- D) -3

8

What is the positive value of x for the following equation?

$$2|x + 4| = 12$$

- A) 0 B) 1 C) 2 D) 3

9

A researcher collected a sample of 12 height samples from his local mall and wrote them down in meters.

The data he collected are 1.23, 1.45, 1.67, 1.53, 1.12, 1.23, 1.25, 1.39, 1.73, 1.84, 1.39, and 1.09.

Which of the following statements is correct when he collects a last-minute sample of 2.23m?

- A) The mean and the range will increase and the median and standard deviation will decrease
- B) The mean, range, and standard deviation will increase and the median remains the same
- C) The mean, range, median, and standard deviation will increase
- D) The mean, range, median and standard deviation will decrease

10

A circle is inscribed in a square. If the square has a side length of 14cm, what is the area of the circle?

- A) 196π
- B) 49π
- C) 14π
- D) 98π

11

What is the new y -intercept of the function $f(x)$ below if the line is moved 2 units to the left and 1 unit up?

$$f(x) = 3x + 5$$

- A) 6 B) 8 C) 12 D) 9

12

Which of the following coordinates is a solution to the following system of inequalities?

$$-2y > -4$$

$$3x - y > 3$$

- A) (1, 2)
B) (1, 3)
C) (2, 4)
D) (3, 1)

13

A circle has endpoints of a diameter with the coordinates of $(-6, 10)$ and $(2, 10)$. Which of the following is an equation for the circle?

- A) $(x - 2)^2 + (y - 10)^2 = 16$
B) $(x + 2)^2 + (y - 10)^2 = 16$
C) $(x + 2)^2 + (y + 10)^2 = 16$
D) $(x - 2)^2 + (y + 10)^2 = 16$

14

If $f(x) = x + 2$ and $g(x) = 2x + 3$, find $f(g(2))$.

- A) 7
B) 4
C) 9
D) 11

15

What is the y -intercept of the function below?

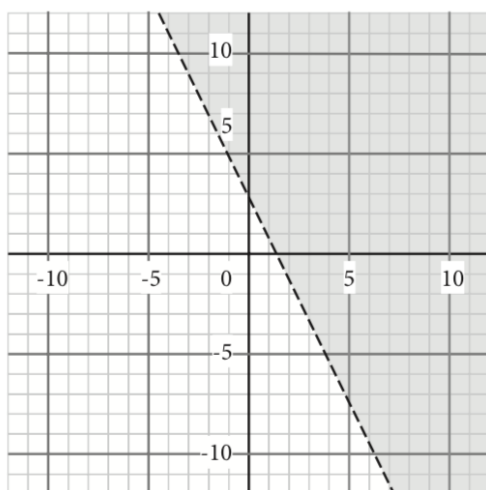
$$y = 300^x + 5$$

- A) 4 B) 5 C) 6 D) 60

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16

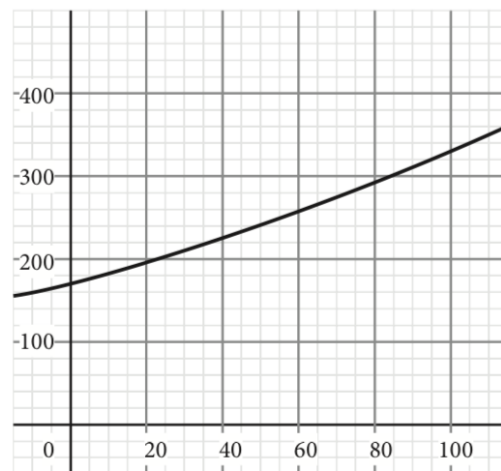
If the shaded region represents the solutions to an inequality, which of the following inequalities is represented by the graph below?



- A) $y > 2x + 3$
- B) $y > -2x + 3$
- C) $y < -x + 3$
- D) $y < -2x + 3$

17

Which of the following expressions is represented by the graph below?



- A) $400(1.03)^{0.2x}$
- B) $180(0.97)^{0.2x}$
- C) $180(1.03)^{0.2x}$
- D) $400(0.97)^{0.2x}$

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18

If $3x + 2(x + 4y) = ax + by$, what is the value of a ?

- A) 3
- B) 2
- C) 4
- D) 5

19

Using interval notation, what is the value of y in the inequality below?

$$-3y > -15$$

- A) $(-\infty, 5)$ B) $(5, \infty)$ C) $(-5; 5)$ D) $\{5\}$

20

If $\sin(3x) = \cos(x + 10)$, what is the value of x in degrees?

- A) 5°
B) 20°
C) 30°
D) 60°

21

Amanda is planning a trip and decides to save for it. She saves \$10 dollars during the first week and doubles the amount she saves every week thereafter. Which type of function below best represents the amount that she will save in a given week?

- A) Exponential growth
B) Linear growth
C) Exponential decay
D) Linear decay

22

If $4x + 8 = 24$, what is the value of x ?

- A) 8
B) 4
C) 3
D) 6

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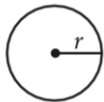
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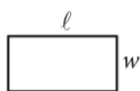
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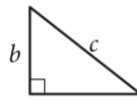
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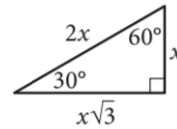
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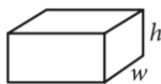
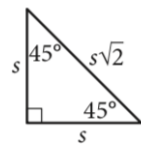
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1

$7x + 28 = 112$, What is the value of $x + 4$?

- A) -4
- B) 16
- C) 8
- D) 12

2

The radius of a ball is three times the radius of its prototype. If the prototype has a volume of 10cm^3 , what is the volume of the ball?

- A) 220 B) 250 C) 270 D) 300

3

If the system of equations below has one solution, what is the value of k ?

$$y = 2x + k$$

$$y = x^2 - 10x + 30$$

- A) -6
- B) 3
- C) 6
- D) -3

4

If $\frac{27^{x-1}}{81^y} = 9$, what is the value of $6x - 8y$?

- A) 2
- B) 5
- C) 4
- D) 10

5

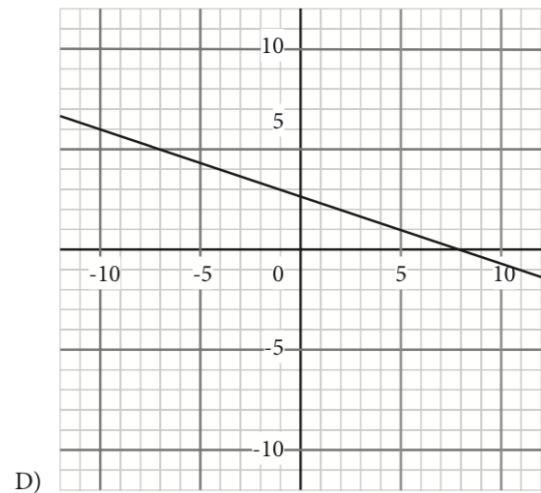
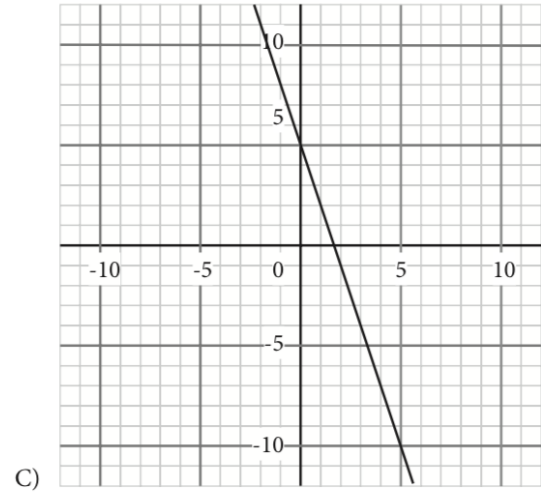
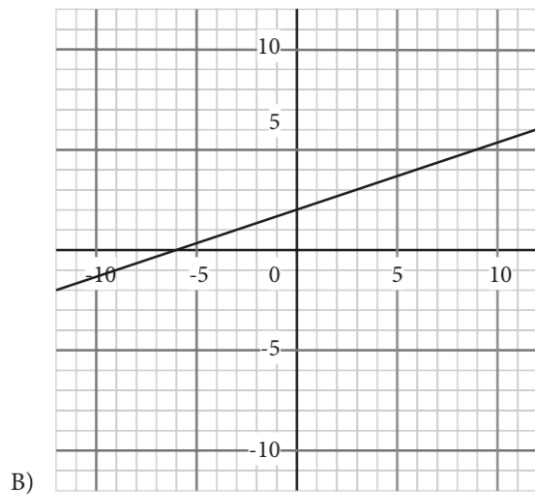
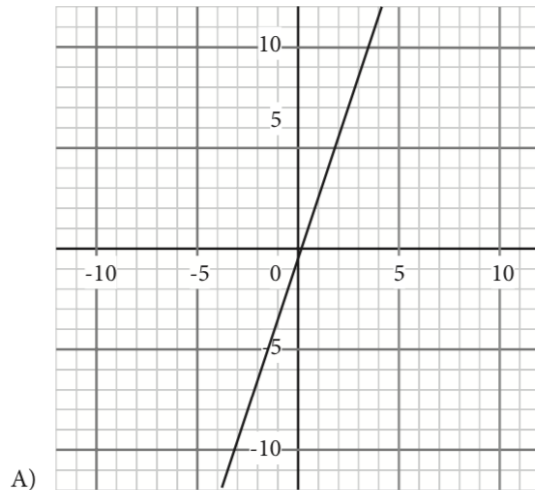
If the diagonal of a square is $16\sqrt{2}$, what is the perimeter of the square?

- A) 16 B) 32 C) 64 D) 128

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6

Which of the following graphs represents the equation $12x + 4y = 8$?



7

If the system of equations below has no solution, what is the value of p ?

$$\frac{3}{2}x + 5 = px + \frac{3}{2}$$

- A) $1/2$ B) 1 C) $3/2$ D) $5/2$

8

A researcher created a model $p(x)$, which is a function for the height of a girl in feet. Where t is in years and t is $2 \leq t \leq 5$. If $p(x) = 1.23(1.32)^t$, which of the following best represents the model in m months?

- A) $p(x) = 1.23(1.32)^{12m}$
B) $p(x) = 1.23\left(\frac{1.32}{12}\right)^m$
C) $p(x) = 1.23(1.32)^{\frac{m}{12}}$
D) $p(x) = 1.23(1.32)^m$

9

If $\frac{-2x^2 + 5x + 10}{x - 3} = A + \frac{7}{x - 3}$, which of the following expressions best represents A ?

- A) $-2x - 1$
B) $2x + 3$
C) $-2x - 3$
D) $-2x^2 + 4x + 13$

10

If $3x = -5y$ and $2x + 3y = -1$, what is the value of x ?

- A) 3
B) -5
C) 2
D) -8

11

What is the greatest possible integer value of k if the quadratic equation below has two distinct solutions?

$$4x^2 - 12x + k = 0$$

- A) 2 B) 4 C) 8 D) 12

12

How many solutions does the equation below have?

$$3(x + 6) + x = 4x + 5$$

- A) 0
B) 1
C) 2
D) Infinite

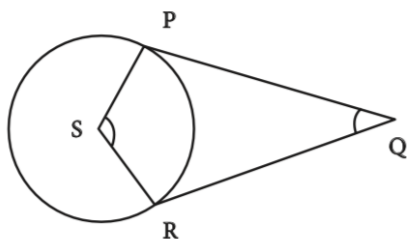
13

Which of the following is equivalent to the expression below?

$$\frac{1}{\frac{1}{x-2} + \frac{1}{x+2}}$$

- A) $\frac{x^2 - 4}{2x}$
 B) x
 C) $\frac{2x}{x^2 - 4}$
 D) $\frac{x^2 + 4}{2x}$

14



PQ and PR are tangents and points P and R lie on the circle. If $\angle PSR = 80^\circ$ and $\angle PQR = a\pi$, what is the value of a ?

- A) $\frac{5}{18}$
 B) $\frac{13}{18}$
 C) $\frac{5}{9}$
 D) 1

15

What is the area of a circle whose circumference is 12π ?

- A) 12π B) 24π C) 36π D) 48π

16

Which of the following expressions is equivalent to the expression below?

$$3p^{\frac{1}{3}}\sqrt{p}$$

- A) $3\sqrt[3]{p}$
 B) $\sqrt[6]{3p^5}$
 C) $3\sqrt[6]{p}$
 D) $3\sqrt[6]{p^5}$

17

What is the volume of a cylinder with a radius of 9 cm and a height of 6 cm ?

- A) 270π
 B) 486π
 C) 81π
 D) 108π

18

If lines $3x + 5y = 6$ and $-6x + py = 18$ are parallel, what is the value of p ?

- A) -10
- B) 5
- C) 10
- D) 3

19

If $3xy(5x + 2) + 2y(3x - 1) = ax^2y + bxy + cy$, what is the value of b ?

- A) 6 B) 9 C) 12 D) 16

20

How many solutions does the absolute value equation below have?

$$-2|x + 4| = -8$$

- A) 0
- B) 1
- C) 2
- D) Infinite

21

If $xy = 5$ and $x^2 + y^2 = 45$, what is the value of $(x + y)^2$?

- A) 9
- B) 55
- C) 110
- D) 35

22

A study was conducted on 1,200 residents in a particular town. 950 of the residents surveyed owned a dog or a cat and the rest owned a different type of pet. If there are 62,500 residents in the town, approximately how many of them own a different type of pet?

- A) 62,500
- B) 49,480
- C) 13,020
- D) 61,550